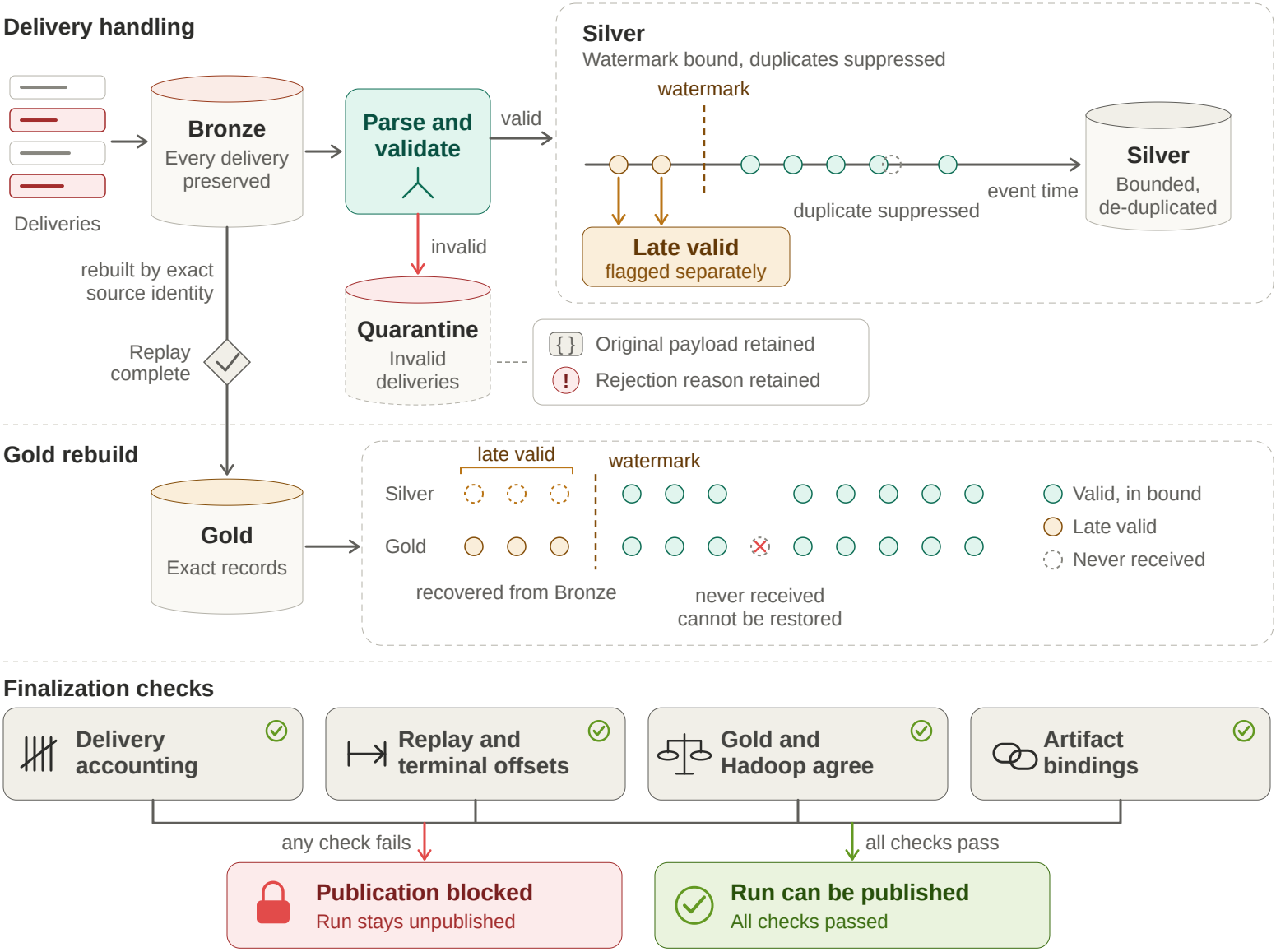

Delivery handling
Deliveries
Bronze
Every delivery preserved
Parse and validate
valid
invalid
Silver
Watermark bound, duplicates suppressed
watermark
event time
duplicate suppressed
Late valid
flagged separately
Silver
Bounded, de-duplicated
rebuilt by exact source identity
Replay complete
Quarantine
Invalid deliveries
Original payload retained
Rejection reason retained
Gold rebuild
Gold
Exact records
late valid
watermark
Silver
Gold
recovered from Bronze
never received
cannot be restored
Valid, in bound
Late valid
Never received
Finalization checks
Delivery accounting
Replay and terminal offsets
Gold and Hadoop agree
Artifact bindings
any check fails
all checks pass
Publication blocked
Run stays unpublished
Run can be published
All checks passed